✈ DGCA CPL / ATPL STUDY NOTES

CHAPTER 6

VHF Direction Finder (VDF)

Radio Navigation & Aids — Ground Training Series

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1. Introduction

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What this section covers: The purpose of VDF, its limitations compared to modern systems, frequency band, and where information on available VDF stations is found.

VHF Direction Finding (VDF) provides a pilot with a **bearing** — the direction toward a ground station. This can be used to:

- **Home** toward the ground station
- **Position fix** — in conjunction with two or more bearings from different stations
- **Track check / position line** — verify or update navigation
- **Let-down procedures** — non-precision approach guidance

Practical Note

VDF is **rarely used** because more sophisticated and accurate systems (VOR, GPS, ILS) are available. However, it requires **no airborne equipment other than a standard VHF radio** — making it available as a backup whenever VHF comms are possible.

Key VDF Facts

- Bearings provided **by voice** on the aircraft's VHF communications frequency
- Frequency range: **118.0 – 137 MHz** (Emission code A3E)
- **Auto-triangulation**: available only on VHF International Distress frequency — **121.5 MHz**
- UHF DF: limited to military use
- Available stations listed in the **AD Section of the AIP**
- Bearings only given when conditions are satisfactory and within calibrated limits

2. Procedures — Q-Codes

What this section covers: The four Q-codes used in VDF, what each provides, and how to request a VDF bearing.

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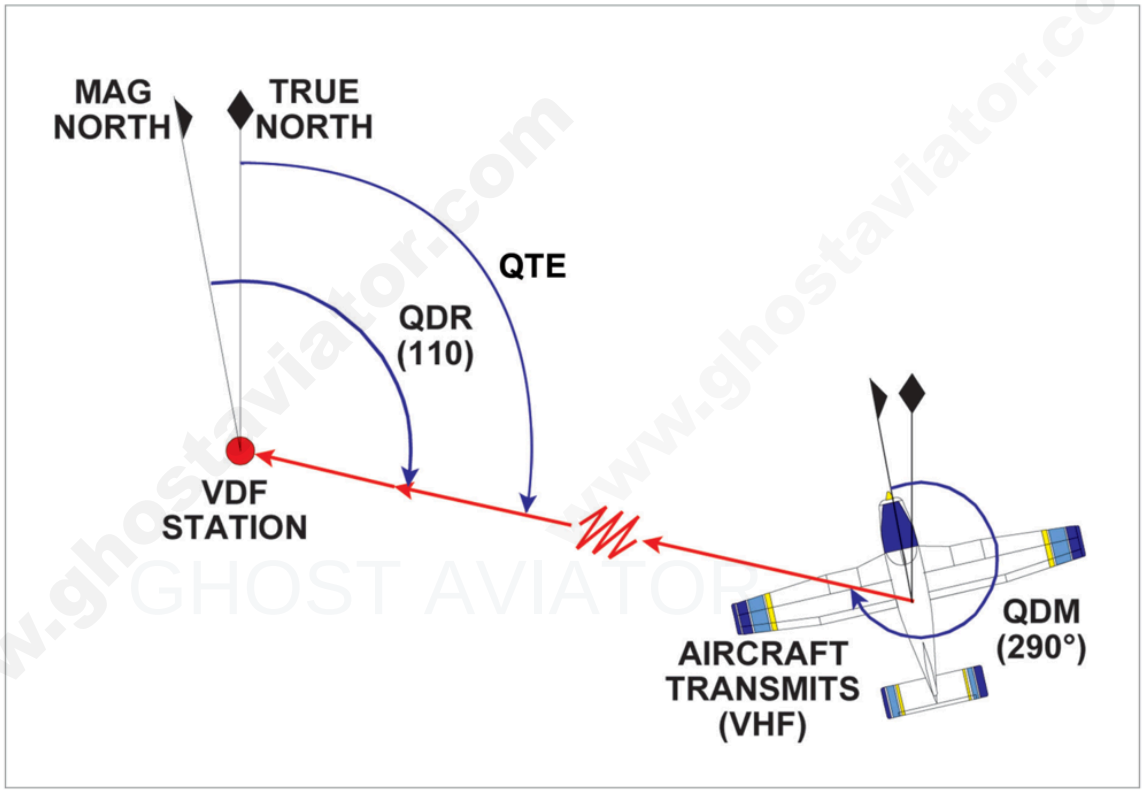


Figure 6.1 QDM/QDR/QTE.

Figure 6.1 — VDF bearing request: the pilot transmits "QDM QDM QDM [station] [callsign] REQUEST QDM". The diagram shows the relationship between the aircraft, station, and the four bearing types.

A pilot requests a VDF bearing using Q-code phrases. The station responds with: the Q-code, the bearing in degrees, the **accuracy class**, and the time of observation (if required).

QDM

Magnetic Heading TO steer to reach station
(assuming zero wind)
Used for: station homing, let-downs

QDR

Magnetic Bearing FROM the station
(the aircraft's radial from the station)
Used for: en route navigation

QTE

True Bearing FROM the station
(the aircraft's true radial from station)
Used for: en route navigation

QUJ

True Track TO the station
(true bearing from aircraft toward station)
Not generally used

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Reciprocal Relationships — Must Know

- **QDM is the reciprocal of QDR** (TO vs FROM; both Magnetic)
- **QUJ is the reciprocal of QTE** (TO vs FROM; both True)
- **QDR, QDM, and QTE are the most commonly used codes**

VDF Q-Codes — Aircraft ↔ Station

QDM	QDR	QTE
Mag heading **TO** station	Mag bearing **FROM** station	True bearing **FROM** station
Aircraft → Station	Station → Aircraft	Station → Aircraft
	(reciprocal of QDM)	

QUJ

True track **TO** station

Aircraft → Station

(reciprocal of QTE)

Converting Between Q-Codes

- **QDR to QDM:** Add or subtract 180° (take the reciprocal)
- **QTE to QDR:** Apply variation — Magnetic = True minus East variation
- **QTE to QUJ:** Add or subtract 180°
- Memory: **"DM = Mag To; DR = Mag From; TE = True From; UJ = True To"**

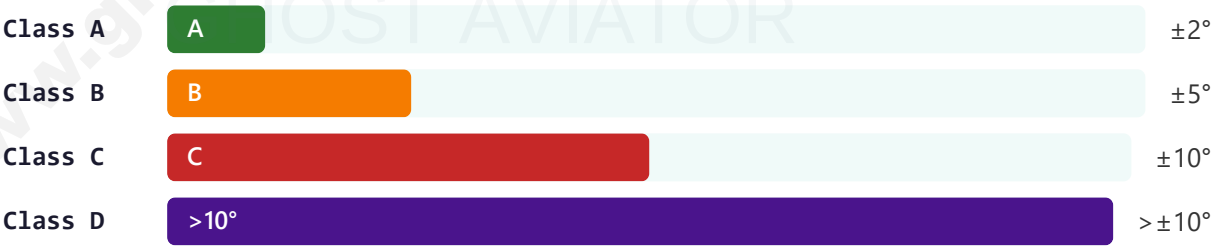
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3. Bearing Accuracy Classes

What this section covers: The four accuracy classes (A–D), what accuracy to assume when no class is given, and the accuracy of modern Doppler-based VDF.

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Class	Accuracy	Notes
Class A	± 2°	Best; rarely achievable in practice
Class B	± 5°	**Normal best case** — bearings normally no better than Class B
Class C	± 10°	Poor conditions
Class D	> ± 10°	Accuracy less than Class C; unreliable
Modern Doppler VDF	± 0.5°	Digital readout; available on VHF and UHF systems



Exam-Critical: Default Accuracy Assumption

> *"Normally, bearings no better than Class B will be available."*

If a VDF bearing is given **without an accuracy classification**, assume **Class B = ±5°**.

4. Principle of Operation

What this section covers: The ground equipment required, the antenna array design, how the bearing is determined and displayed.

The only equipment required **in the aircraft** is a standard VHF radio. On the ground, the VDF station uses a **direction-finding aerial** (circular array of vertical elements) and a display.

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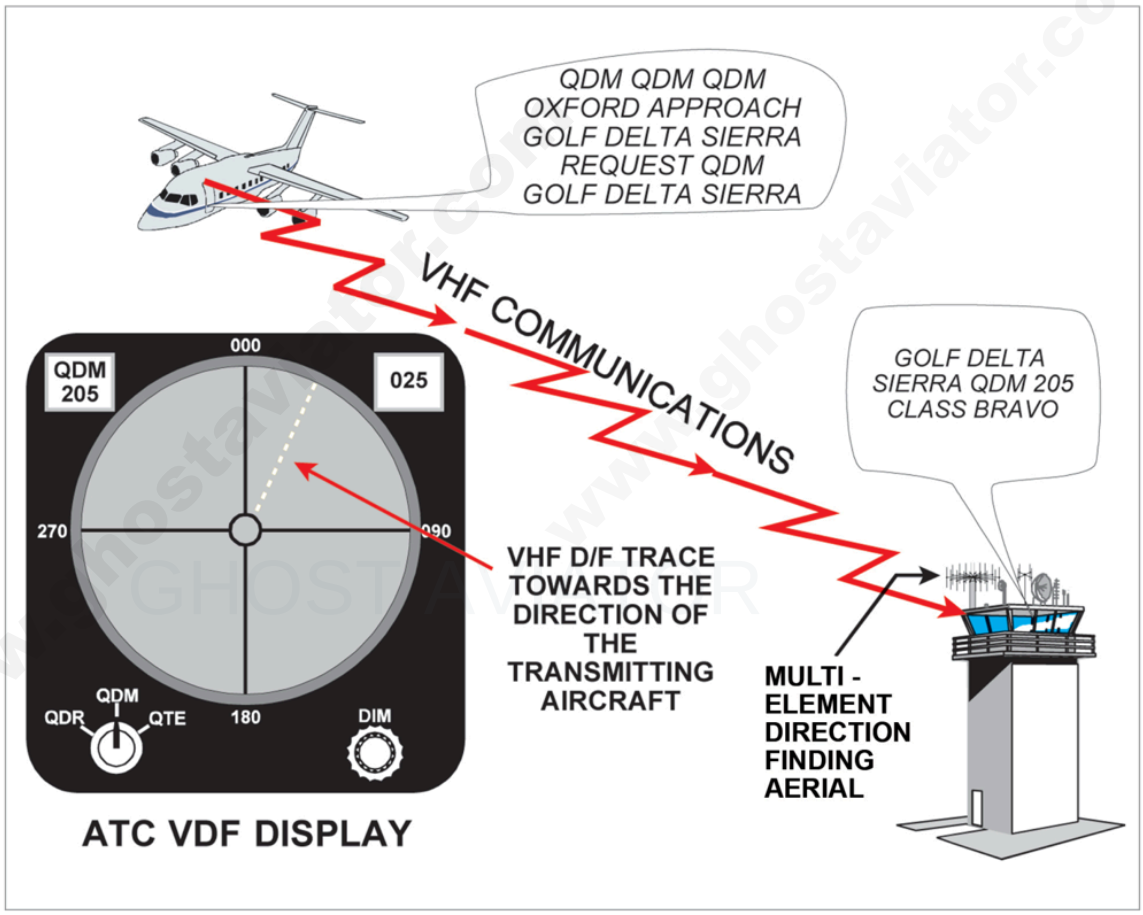


Figure 6.2: Obtaining a VDF Bearing

Figure 6.2 — Obtaining a VDF bearing: the ground station uses a circular array of vertical elements. The VHF transmission from the aircraft is vertically polarized; the array resolves the direction of the incoming signal, displayed on the ground-station screen.

How VDF Works

1. Aircraft transmits on VHF comm frequency — **vertically polarized** signal
2. Ground antenna: circular array of **vertical elements** — matched polarization
3. Equipment resolves bearing from transmissions received at each element in the array
4. Bearing displayed relative to **True or Magnetic North** at the station
5. Latest Doppler-based systems produce a **digital bearing at ±0.5°**

5. Range of VDF

What this section covers: Why VDF range follows the LOS formula, factors that extend or

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reduce range, and how to calculate minimum aircraft altitude or maximum range.

VDF operates in the **VHF band** — therefore range obeys the **line-of-sight formula**:

VDF RANGE (LINE OF SIGHT)

$$\text{Range (NM)} = 1.25 \times (\sqrt{h_{TX}} + \sqrt{h_{RX}})$$

h_{TX} = transmitter height in feet | h_{RX} = receiver height in feet

Worked Examples — VDF Range Calculations

Q1 (find min aircraft altitude): Station = 2500 ft, Range needed = 300 NM

$300 = 1.25 \times (\sqrt{2500} + \sqrt{h})$

$300 / 1.25 = 240 = 50 + \sqrt{h}$

$\sqrt{h} = 240 - 50 = 190$

$h = 190^2 =$ **36,100 ft** ✓ (answer c)

Q4 (find max range): Aircraft = 9000 ft, Station = 400 ft

$\text{Range} = 1.25 \times (\sqrt{9000} + \sqrt{400}) = 1.25 \times (94.87 + 20)$

$= 1.25 \times 114.87 =$ **143.6 NM ≈ 143 NM** ✓ (answer c)

Q6 (find max range): Aircraft = 19,000 ft, Station = 1400 ft

$\text{Range} = 1.25 \times (\sqrt{19000} + \sqrt{1400}) = 1.25 \times (137.84 + 37.42)$

$= 1.25 \times 175.26 =$ **219.1 NM ≈ 219 NM** ✓ (answer d)

Factors Affecting VDF Range

- **Altitude of aircraft and station** (primary factor) — higher = greater range
- **Intervening high ground** — limits range, especially for low-flying aircraft in hilly terrain
- **Transmitter power** — both airborne and ground
- **Atmospheric ducting** (super-refraction) — can give greater than LOS range

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6. Factors Affecting Accuracy

> **What this section covers:** The five factors that reduce VDF bearing accuracy.

Factor	Effect on Accuracy
Propagation error / Site error	Aircraft transmissions reflected from terrain or buildings at the site, arriving at the antenna from false directions — producing bearing errors
Aircraft attitude	VHF comms are vertically polarized; best results when aircraft flies **straight and level**. Bank angle tilts the antenna axis, reducing bearing accuracy
Directly overhead effect	Poor accuracy when aircraft is directly overhead the VDF station — especially with modern Doppler systems. Direct and ground-reflected waves arrive simultaneously causing fading/loss
Multi-path signals	Direct + reflected waves reaching the antenna simultaneously → fading and bearing errors. Usually short-lived as aircraft transits the area
Simultaneous transmissions	Two or more aircraft transmitting at the same time → momentary bearing errors

7. Determination of Position (Auto-triangulation)

> **What this section covers:** How position fixes are obtained from multiple VDF bearings and the auto-triangulation service.

If **multiple ground stations** are available and linked to an ATCC, the aircraft's position can be fixed using **auto-triangulation** — bearings from different stations are automatically crossed to give a position fix, transmitted to the pilot.

> **Auto-triangulation Limitation**
>
> Auto-triangulation may be available to **Distress and Diversion (D&D) Cells** — but **cannot be guaranteed**. It is available only on the VHF International Distress Frequency: **121.5 MHz**.

8. VDF Summary

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Bearings:	QDM - Mag TO Station QDR - Mag FROM Station QUJ - True TO Station QTE - True FROM Station
Uses:	Track Check Position Line Homing Let-downs
Class:	A = ± 2° B = ± 5° C = ± 10° D = > 10°
Principle:	Ground Equipment – Direction Finding Aerial CRT Display
Range:	Line of Sight Power of Transmitters Intervening High Ground Atmospheric Conditions (Ducting)
Accuracy:	Propagation Error Site Error Aircraft Attitude Overhead Fading Due to Multi-path Signals
Position Service:	Position Fixing by Auto-triangulation

Figure 6.3 VDF Summary

Figure 6.3 — VDF Summary reference: consolidates Q-codes, uses, accuracy classes, principle, range, and accuracy factors in a single-page reference.

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QUICK REVISION SUMMARY — CHAPTER 6

- **QDM** = Mag heading TO station (for homing/let-down)
- **QDR** = Mag bearing FROM station (reciprocal of QDM)
- **QTE** = True bearing FROM station
- **QUJ** = True track TO station (reciprocal of QTE; rarely used)
- Class A=±2°, B=±5°, C=±10°, D=>10°
- **No class stated → assume Class B = ±5°**
- Modern Doppler VDF = ±0.5°
- VHF band: 118.0–137 MHz; A3E emission; vertically polarized
- LOS formula: **Range = 1.25 × (√hTX + √hRX)**
- Auto-triangulation: 121.5 MHz only; not guaranteed
- No airborne equipment needed other than a standard VHF radio
- Accuracy degraded by: propagation error, site error, aircraft attitude, overhead effect, multi-path, simultaneous transmissions

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Practice Questions & Detailed Answers

Q1. An aircraft has to communicate with a VHF station at a range of 300 NM. If the ground station is situated 2500 ft AMSL, which of the following is the lowest altitude at which contact is likely to be made?

a. 190 ft

b. 1,378 ft

c. 36,100 ft

d. 84,100 ft

Correct Answer: **(c)** — 36,100 ft

Working:

$300 = 1.25 \times (\sqrt{2500} + \sqrt{h})$

$300/1.25 = 240 = 50 + \sqrt{h}$

$\sqrt{h} = 190 \rightarrow h = 190^2 =$ **36,100 ft**

Why the other options are wrong:

- **(a) 190 ft** — confuses $\sqrt{h} = 190$ with $h = 190$. Never forget to square the result.
- **(b) 1,378 ft** — no valid derivation; likely a misapplied partial formula.
- **(d) 84,100 ft** — result of $290^2 = 84{,}100$; obtained by adding $\sqrt{h}$_station (50) to 240 instead of subtracting: 240+50=290. Addition trap.

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Instructor's Note: *The most common error is option (a) — computing √h = 190 and stopping there. Option (d) is the "addition instead of subtraction" trap. Always: isolate √h by subtracting the station term, then square to get h.*

Q2. Class 'B' VHF DF bearings are accurate to within:

a. ± 1°

b. ± 5°

c. ± 2°

d. ± 10°

Correct Answer: **(b)** — ± 5°

Explanation: Class B = ±5°. The four classes: A = ±2°, B = ±5°, C = ±10°, D = >±10°.

Why the other options are wrong:

- **(a) ±1°** — better than Class A; not a defined VDF class (only modern Doppler achieves ±0.5°).
- **(c) ±2°** — this is Class A, not Class B. Classic swap between A and B.
- **(d) ±10°** — this is Class C. Another class confusion trap.

Instructor's Note: *Most common wrong answer is (c) — swapping Class A (±2°) and Class B (±5°). Remember: A=2, B=5, C=10. Class B (±5°) is the normal best case in practice.*

Q3. A VDF QDM given without an accuracy classification may be assumed to be accurate to within:

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a. 2 degrees

b. 5 degrees

c. 7.5 degrees

d. 10 degrees

Correct Answer: **(b)** — 5 degrees

Explanation: The rule states: "*Normally, bearings no better than Class B will be available.*" Therefore, when no accuracy class is specified, the bearing should be assumed to be **Class B = ±5°**.

Why the other options are wrong:

- **(a) 2°** — Class A: too optimistic; not normally achievable with conventional VDF.
- **(c) 7.5°** — between Class B and C; not a defined class. Made-up distractor.
- **(d) 10°** — Class C: too pessimistic. The text says "no better than Class B" meaning Class B is the assumed worst-normal case, not Class C.

Instructor's Note: *This tests the "default assumption" rule. "Bearings no better than Class B" means Class B (±5°) is the default when no class is stated. This is directly examinable.*

Q4. An aircraft at altitude 9000 ft wishes to communicate with a VHF/DF station situated at 400 ft AMSL. What is the maximum range at which contact is likely to be made?

a. 115 NM

b. 400 NM

c. 143 NM

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d. 63.5 NM

Correct Answer: **(c)** — 143 NM

Working:
Range = 1.25 × (√9000 + √400) = 1.25 × (94.87 + 20)
= 1.25 × 114.87 = **143.6 ≈ 143 NM**

Why the other options are wrong:

- **(a) 115 NM** — possibly using only √9000 component, or using coefficient 1.23 instead of 1.25: 1.23 × 114.87 ≈ 141 NM. Neither gives 115 NM exactly; likely a wrong formula path.
- **(b) 400 NM** — unrelated to the calculation; likely confused with the station elevation figure of 400 ft.
- **(d) 63.5 NM** — result of using only the station component: 1.25 × 20 × 2 or similar incomplete calculation.

Instructor's Note: *√9000 ≈ 94.87 (use a calculator; this is not a round number). √400 = 20 exactly. Sum = 114.87. × 1.25 = 143.6 NM. Round to 143 NM.*

Q5. An aircraft is passed a true bearing from a VDF station of 353°. If variation is 8°E and the bearing is classified as 'B' then the:

a. QDM is 345° ± 5°

b. QDR is 345° ± 2°

c. QTE is 353° ± 5°

d. QUJ is 353° ± 2°

Correct Answer: **(c)** — QTE is 353° ± 5°

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Explanation:

A "true bearing FROM the station" = **QTE** = 353°

Class B accuracy = **±5°**

Therefore: **QTE = 353° ± 5°**

Deriving all Q-codes for reference:

QTE (True FROM) = 353° (given)

QDR (Mag FROM) = 353° − 8°E variation = **345°**

QUJ (True TO) = 353° − 180° = **173°**

QDM (Mag TO) = 173° − 8° = **165°**

Why the other options are wrong:

- **(a) QDM = 345°** — 345° is the value of QDR (mag FROM), not QDM. QDM = 165°. Wrong Q-code identified.
- **(b) QDR = 345° ± 2°** — 345° is correct for QDR, but Class B accuracy is ±5°, not ±2° (that is Class A). Correct bearing, wrong accuracy class.
- **(d) QUJ = 353° ± 2°** — QUJ is the True track TO station = 173° (reciprocal of QTE), not 353°. Also ±2° is Class A not Class B. Two errors in one option.

Instructor's Note: *This is the most complex question in the chapter. Three traps: Q-code identity, correct bearing value, correct accuracy class. Work through: (1) True bearing FROM = QTE; (2) Class B = ±5°. The most instructive wrong answer is (b) — it has the right bearing for QDR but the wrong class; a careful student should spot both must be correct.*

Q6. An aircraft at 19,000 ft wishes to communicate with a VDF station at 1400 ft AMSL. What is the maximum range at which contact is likely?

a. 175 NM

b. 400 NM

c. 62.5 NM

d. 219 NM

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Correct Answer: **(d)** — 219 NM

Working:

Range = 1.25 × (√19000 + √1400) = 1.25 × (137.84 + 37.42)

= 1.25 × 175.26 = **219.1 ≈ 219 NM**

Why the other options are wrong:

- **(a) 175 NM** — this is the sum of √19000 + √1400 ≈ 175.26, before multiplying by 1.25. Forgetting the final multiplication step. Extremely common error.
- **(b) 400 NM** — not derivable from correct calculation; red herring.
- **(c) 62.5 NM** — possibly using only the station height term: 1.25 × √1400 × 2 ≈ 93.6 NM; no clear correct derivation.

Instructor's Note: *Option (a) is the classic "forgot to multiply by 1.25" trap — the sum of square roots is 175.26, which exactly matches option (a). Always complete the full formula. √19000 ≈ 137.84; √1400 ≈ 37.42.*

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Master Reference Tables

Q-Code Quick Reference

Code	Meaning	True/Mag	To/From	Primary Use
QDM	Magnetic Heading TO steer to station	Magnetic	TO	Homing, let-downs
QDR	Magnetic Bearing FROM station (radial)	Magnetic	FROM	En route navigation
QTE	True Bearing FROM station	True	FROM	En route navigation
QUJ	True Track TO station	True	TO	Rarely used

Accuracy Classes

Class	Accuracy	Default?
A	±2°	No — rarely achievable
B	±5°	**YES — default when no class given**
C	±10°	No
D	>±10°	No
Modern Doppler	±0.5°	Specific to Doppler systems

LOS Calculation Reference

Q	Given Heights	Calculation	Answer
Q1	Station=2500 ft; Range=300 NM	300/1.25=240; 240−50=190; 190^2	**36,100 ft**
Q4	AC=9000 ft; Stn=400 ft	1.25×(94.87+20)=1.25×114.87	**143 NM**
Q6	AC=19,000 ft; Stn=1400 ft	1.25×(137.84+37.42)=1.25×175.26	**219 NM**

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Answer Key — Chapter 6

Q1	Q2	Q3	Q4
c	b	b	c

Q5	Q6
c	d

Mnemonics

- **QDM:** M = Mag TO station (use to Home; M like Magnetic, TO like steer-To)
- **QDR:** R = Radial FROM station (Magnetic)
- **QTE:** E = truE bearing FROM station
- **QUJ:** True track TO station; J = just rarely used
- **Reciprocals:** DM ↔ DR (both Mag, one TO one FROM); TE ↔ UJ (both True)
- **Class A=2°, B=5°, C=10°, D=dunno**
- **No class = Class B = ±5°**
- **LOS: 1.25 × ($\sqrt{h_1}$ + $\sqrt{h_2}$)** — heights in feet, answer in NM
- **Auto-triangulation = 121.5 MHz only**

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